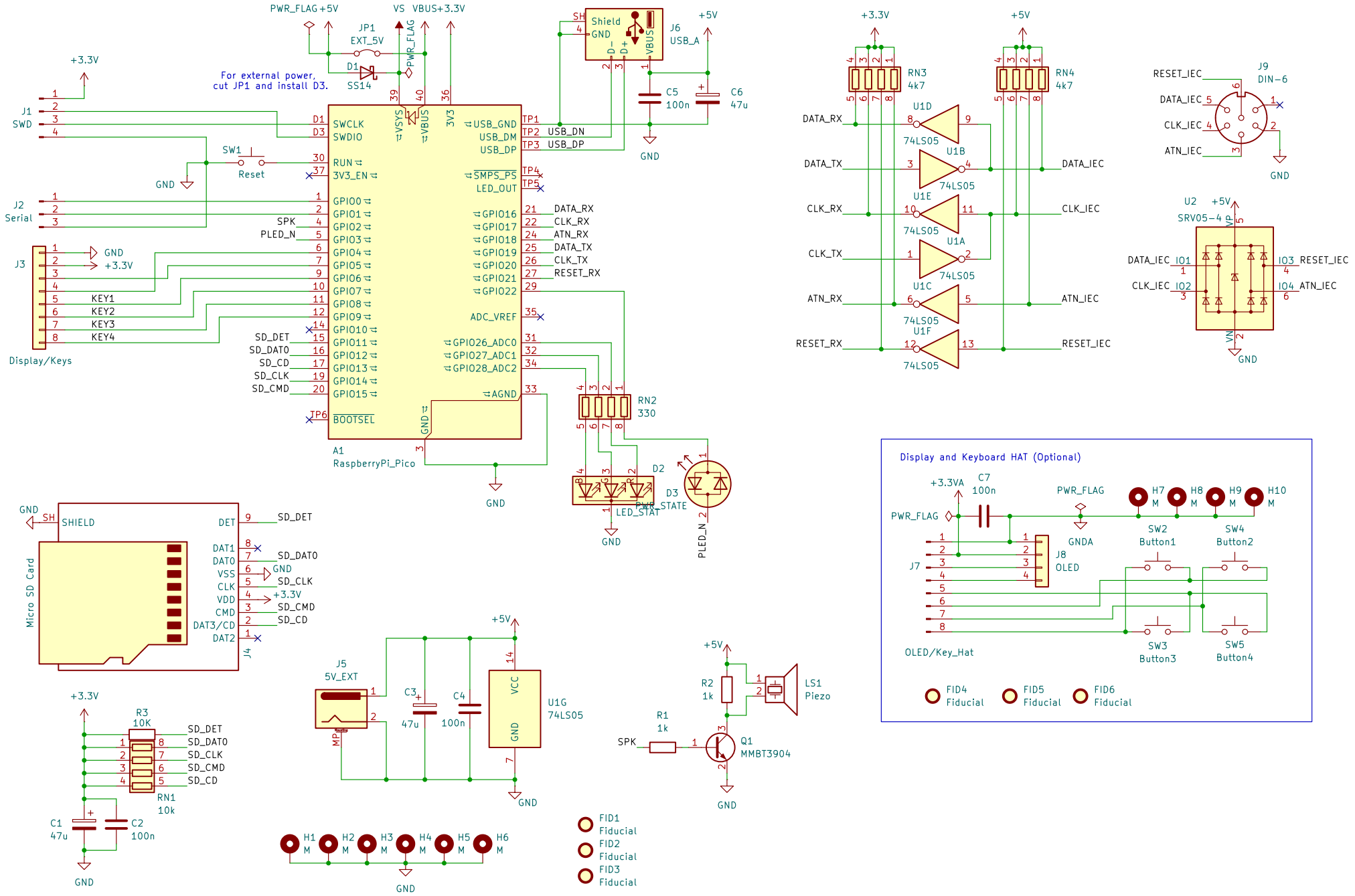

PWR_FLAG +5V
VS VBUS +3.3V
JP1
EXT_5V
D1
SS14
For external power,
cut JP1 and install D3.
Shield
GND
J6
USB_A
+5V
C5
100n
C6
47u
+3.3V
J1
SWD
SWCLK
SWDIO
SW1
Reset
RUN
3V3_EN
GND
J2
Serial
SPK
PLED_N
J3
Display/Keys
KEY1
KEY2
KEY3
KEY4
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
SD_DET
SD_DAT0
SD_CD
SD_CLK
SD_CMD
BOOTSEL
TP6
VSYS
VBUS
3V3
USB_GND
USB_DM
USB_DP
USB_DN
TP1
TP2
TP3
SMPS_PS
LED_OUT
TP4
TP5
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
ADC_VREF
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
AGND
DATA_RX
CLK_RX
ATN_RX
DATA_TX
CLK_TX
RESET_RX
A1
RaspberryPi_Pico
RN2
330
D2
LED_STAT
D3
PWR_STATE
PLED_N
+3.3V
RN3
4k7
+5V
RN4
4k7
U1D
U1B
U1E
U1A
U1C
U1F
74LS05
DATA_RX
DATA_TX
CLK_RX
CLK_TX
ATN_RX
RESET_RX
DATA_IEC
CLK_IEC
ATN_IEC
RESET_IEC
J9
DIN-6
RESET_IEC
DATA_IEC
CLK_IEC
ATN_IEC
GND
U2
+5V
SRV05-4
VP
VN
IO1
IO2
IO3
IO4
DATA_IEC
CLK_IEC
RESET_IEC
ATN_IEC
GND
Micro SD Card
SHIELD
DET
DAT1
DAT0
VSS
CLK
VDD
CMD
DAT3/CD
DAT2
SD_DET
SD_DAT0
GND
SD_CLK
+3.3V
SD_CMD
SD_CD
J4
+3.3V
R3
10K
RN1
10k
C1
47u
C2
100n
GND
J5
5V_EXT
MP
C3
47u
C4
100n
+5V
U1G
74LS05
VCC
GND
GND
R2
1k
LS1
Piezo
R1
1k
SPK
Q1
MMBT3904
GND
H1
H2
H3
H4
H5
H6
M
GND
FID1
FID2
FID3
Fiducial
Display and Keyboard HAT (Optional)
+3.3VA
C7
100n
PWR_FLAG
GNDA
H7
H8
H9
H10
M
SW2
Button1
SW4
Button2
SW3
Button3
SW5
Button4
J7
J8
OLED
OLED/Key_Hat
FID4
FID5
FID6
Fiducial